A. Numbers

time limit per test: 1 second
memory limit per test: 64 megabytes

Little Petya likes numbers a lot. He found that number 123 in base 16 consists of two digits: the first is 7 and the second is 11. So the sum of digits of 123 in base 16 is equal to 18.

Now he wonders what is an average value of sum of digits of the number A written in all bases from 2 to $A - 1$.

Note that all computations should be done in base 10. You should find the result as an irreducible fraction, written in base 10.

Input

Input contains one integer number A ($3 \le A \le 1000$).

Output

Output should contain required average value in format «X/Y», where X is the numerator and Y is the denominator.

Examples

input Copy

```
5
```

output Copy

```
7/3
```

input Copy

```
3
```

output Copy

```
2/1
```

Note

In the first sample number 5 written in all bases from 2 to 4 looks so: 101, 12, 11. Sums of digits are 2, 3 and 2, respectively.

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 13
Finished
Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
369975796	Apr/06/2026 15:45	Accepted
369975554	Apr/06/2026 15:43	Compilation error
369975246	Apr/06/2026 15:41	Compilation error

→ Problem tags

implementation math *1000

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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